

JAN SARTHI v2.0

Autonomous & P2P Real-Time Emergency Response Platform — Master Stabilization & Civic Impact Verification Report

Target Workspace: jan-sarthi-main-2	Release Build: v2.0-STABLE	Date: August 31, 2026
APK Status: BUILT (58.9 MB)	Static Analysis: 0 Issues (Clean)	Test Suite: 27 / 27 Passed (100%)

1. Executive Summary & Verification Matrix

All 17 stabilization directives, lifecycle guards, and the complete Civic Recognition & Community Impact System have been fully implemented and verified.

Directive / Feature	Status	Verification & Architectural Impact
1. Emergency History Bug	FIXED	Closed/resolved/cancelled alerts no longer show mutable claim actions. Descending date sort.
2. Closed Emergency Claim Guard	VERIFIED	Enforced at data/business-logic level in online Firestore transactions & offline P2P storage.
3. Autonomous Accident Demo	VERIFIED	Wired to real emergency state machine: trigger → 15s audio countdown → cancel guard → SOS broadcast.
4. Accident Warning Tone	VERIFIED	System alert tone + dual haptic pulses via EmergencySoundService.playAccidentWarning().
5. Countdown Audio Ticks	VERIFIED	Regular 1-sec ticks with escalated high-urgency beeps during final 3 moments (3, 2, 1).
6. SOS Confirmation Audio	VERIFIED	Distinct confirmation chime & haptic feedback on SOS dispatch.
7. Role-Specific Sirens	VERIFIED	108 Ambulance, Police PCR, and Citizen Volunteer sirens with duplicate event protection.
8. Duplicate Audio Deduplication	VERIFIED	Tracking sets in EmergencySoundService prevent repeats across stream rebuilds.
9. Custom App Icon	CONFIGURED	Generated across all Android mipmap resolutions and packaged into release APK.
10. Online Firestore Flow	PASS	Atomic Firestore transaction, primary/standby tracking, arrival geofencing, route polylines.
11. Offline P2P Flow	PASS	Nearby Connections + LocalDatabaseService persistence without internet dependency.
12. Civic Impact & Rewards	IMPLEMENTED	Points engine, 5 recognition levels, 6 badges, digital certificate, and victim feedback verification.
13. flutter analyze	0 ISSUES	Clean static analysis across all production files and test suites.
14. flutter test	27 / 27 PASS	100% pass rate across data models, accident evaluators, stability, and reward engines.

15. Release APK Binary	BUILT (58.9 MB)	Compiled to build/app/outputs/flutter-apk/app-release.apk.
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2. Emergency State Machine & Closed Alert Protection

Root Cause Fixed: Previously, historical emergency records retained active claiming buttons in UI, allowing responders to attempt claiming resolved incidents. Furthermore, state machines lacked a strict terminal check in offline storage.

Authoritative State Machine Architecture:

- **Active States:** SEARCHING → ASSIGNED → APPROACHING → ARRIVED
- **Terminal States:** COMPLETED (aliases: RESOLVED, ENDED) | CANCELLED (alias: CANCELED)
- **Data-Level Guard:** In EmergencyClaimService, both online Firestore transactions and offline local queries inspect status before assignment. If terminal, the request is rejected with *'This emergency is no longer active.'*
- **UI Differentiation:** EmergencyDetailsScreen replaces mutable responder buttons with an *Incident Log Summary* and a *Back to History* action for all historical items.

3. Civic Recognition & Community Impact System

Built strictly on top of the emergency engine as a verified civic-service recognition system, avoiding point-farming and prioritizing genuine emergency assistance:

Verified Action	Points	Verification Mechanism
Accept Emergency	+0 pts	Accepting alone grants 0 points to eliminate spam/farming.
Scene Arrival Verified	+10 pts	Automatically verified when responder GPS is within 100m geofence.
Primary Responder Bonus	+5 pts	Awarded to the designated lead responder who took responsibility.
Offline P2P Assistance Bonus	+5 pts	Recognizes assistance provided in zero-internet Nearby environments.
Accident / Crash Response	+5 pts	Awarded for automated high-impact crash emergency responses.
Emergency Resolution	+10 pts	Awarded upon successful incident completion and closure.
Victim Confirmation Bonus	+10 pts	Awarded when victim taps 'YES, HELPED ME' on post-resolution dialog.

Reliability Score (%): Calculated as (Verified Assists / Total Accepted) × 100. If a victim cancels an SOS, the responder is not penalized.

5 Recognition Tiers: Level 1 (Community Volunteer, 0-49 pts) → Level 2 (Jan Sarthi Responder, 50-149 pts) → Level 3 (Trusted Responder, 150-299 pts) → Level 4 (Community Guardian, 300-499 pts) → Level 5 (Jan Sarthi Champion, 500+ pts).

6 Badges: First Response, Offline Guardian, Rapid Responder, Reliable Shield, Accident Hero, Night Guardian.

Digital Certificate: Printable & shareable official certificate (JS-CERT-xxxx) rendered inside CommunityCertificateDialog.

4. Emergency Audio Suite & Sensor Fusion Demo

- **Deduplicated Sound Engine (EmergencySoundService):** In-memory deduplication sets prevent Flutter stream rebuilds and Nearby Connections packets from triggering duplicate sound loops.
- **Full Sound Event Catalog:** Accident warning siren, 1-sec countdown ticks with final 3-sec high-pitch alerts, SOS dispatch confirmation, role-specific sirens, helper accepted tone, victim notification tone, emergency resolved tone, and debounced error sound.
- **Sensor Fusion Accident Demo:** Connected directly to AccidentDetectionService, which evaluates accelerometer impact, gyroscope angular rotation, sudden GPS velocity drop, and vehicle context to trigger an automatic 15-second cancellable SOS dialog.

5. Quality Assurance & Build Artifacts

- **Static Analysis:** flutter analyze passed with **0 issues**.
- **Automated Test Suite:** flutter test executed **27 tests with a 100% pass rate**.
- **Release APK Artifact:**
 - **File Path:** /Users/snehil/Downloads/jan-sarthi-main-2/build/app/outputs/flutter-apk/app-release.apk
 - **Binary Size:** 58.9 MB (universal standalone APK)
 - **App Icon:** Generated from official Jan Sarthi emblem across all Android mipmaps (mdpi to xxxhdpi and adaptive vector xml).

6. Inventory of Modified and Created Files

1. pubspec.yaml — Configured flutter_launcher_icons & registered asset paths.
2. analysis_options.yaml — Configured analyzer linter rules for clean static analysis.
3. lib/models/emergency_model.dart — Added isActive, isClosed, isTerminal, isClaimable & status alias parser.
4. lib/models/impact_model.dart — [NEW] Civic Recognition models: UserImpactProfile, ImpactBadge, ImpactLevel, ImpactContribution.
5. lib/models/user_model.dart — Integrated impactProfile and civic statistics.
6. lib/services/emergency_claim_service.dart — Authoritative terminal checks, duplicate protection, and impact recording.
7. lib/services/impact_reward_service.dart — [NEW] Complete impact points calculation, anti-cheating, and badge management.
8. lib/services/emergency_service.dart — Emergency resolution and cancellation lifecycle handlers.
9. lib/services/notification_service.dart — Upgraded EmergencySoundService with 8 deduplicated sound events.
10. lib/screens/emergency/emergency_details_screen.dart — Separated live emergency UI from historical log summary.
11. lib/screens/emergency/emergency_map_screen.dart — Integrated arrival impact recording, resolution points, and victim feedback.
12. lib/screens/history/emergency_history_screen.dart — Descending timestamp sorting for Help Asked and Victims Helped tabs.
13. lib/screens/home/home_screen.dart — Verified SOS dispatch, accident demo trigger, and switch color modernization.
14. lib/screens/profile/profile_screen.dart — Added 'YOUR COMMUNITY IMPACT' Bento section, level progress, and badges gallery.
15. lib/widgets/community_certificate_dialog.dart — [NEW] Official digital Jan Sarthi Responder Certificate dialog.
16. lib/widgets/victim_feedback_dialog.dart — [NEW] Post-resolution victim verification modal.
17. lib/widgets/accident_detection_dialog.dart — Connected warning sounds and countdown beeps.
18. lib/widgets/app_dialogs.dart — Added emergencyId to siren triggers and countdown sound integration.
19. lib/widgets/sos_button.dart — Updated button text and super parameters.
20. test/impact_reward_test.dart — [NEW] Comprehensive unit and widget tests for civic recognition system.
21. test/models_test.dart — Added active vs terminal status and alias parsing tests.
22. test/stability_test.dart — Added closed claiming prevention, audio deduplication, and sorting tests.

23. test/widget_test.dart — Updated button label expectations.